

Closed-Loop Reveals What Open-Loop Hides: Generation-Coherence Collapse and Group-Wise Recovery in Quantized Autonomous-Driving VLAs

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Abstract—Post-training quantization of autonomous-driving vision-language-action (VLA) models is widely reported lossless on open-loop nuScenes L2. We show this is a metric artifact and that the truth only appears closed-loop. On OpenDriveVLA-0.5B (Qwen2.5-0.5B planner on UniAD perception) we establish three results. (1) Open-loop is blind: INT8/INT4 weight quantization is lossless (0.33 m), but zeroing the ego-state input inflates L2 to 6.38 m (19 $\times$)—open-loop L2 is $\approx$ ego extrapolation and cannot see perception/planning or quantization damage. (2) Closed-loop reveals a failure: in a real-camera neural-rendering simulator (NeuroNCAP, 14 scenes / 16 scenario instances), naive per-channel INT4 collapses generation—degenerate trajectory text on nearly every frame, planner frozen ($\leq 15\%$ well-formed on 15/16, mean 3.7%)—while FP16 stays coherent (mean 76.0%). (3) Granularity is the fix: group-wise INT4 (group 128, ≈ 0.1 extra bits/weight) recovers most of FP16’s coherence (mean 73.5%, $\geq 85\%$ of FP16 on 13/16). The collapse is a quantization $\times$ deployment-domain-shift interaction (per-channel INT4 is lossless in-distribution), governed by scale granularity, not bit-width. We argue generation-coherence rate is the correct metric (collision counts are a confounded proxy) and that group-wise INT4 is the actionable, nearly-free recommendation. We are explicit about scope: results are two-scene with an out-of-distribution renderer (one scene is failed even by FP16), establishing the mechanism rather than a fleet-wide guarantee.

Index Terms—vision-language-action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

On-board inference motivates quantizing driving VLAs, and 4-bit weight quantization is generally reported free on open-loop nuScenes planning. We show that conclusion cannot be drawn from open-loop L2, and that closed-loop evaluation exposes a real, fixable failure. Contributions:

- A controlled demonstration that open-loop nuScenes L2 is $\approx$ ego-motion extrapolation (19 $\times$ ego-ablation), hence structurally blind to quantization damage.
- The first closed-loop characterization of weight-quantization effects in a driving VLA, showing per-channel INT4 collapses generation under deployment-domain shift while group-wise INT4 does not.

- Generation-coherence rate as the un-confounded metric, and a near-free deployment recommendation (group-wise INT4), with component memory benchmarks toward edge deployment.

II. Background

OpenDriveVLA pairs a Qwen2.5 language/action head with UniAD stage-1 track+map perception (a ResNet-101 multi-camera backbone + temporal BEVFormer) as a vision tower; the planner emits a trajectory as text. Open-loop nuScenes evaluation reports L2 displacement against logged futures; prior work (“ego status is all you need”) noted it is dominated by ego state. NeuroNCAP is a photorealistic closed-loop simulator: a NeuRAD neural renderer re-renders sensor views at the ego’s driven pose, so the logged trajectory cannot be coasted on. Group-wise quantization assigns a scale per block of weights rather than per channel; it is standard for INT4 LLMs because it limits the damage of weight outliers on the shared scale.

III. Method

We apply weight-only post-training quantization (round-to-nearest, dequantized to the compute dtype) to the 168 Linear layers of the Qwen-0.5B backbone (357.8 M params); perception, projectors, embeddings, and head remain FP, isolating the language head’s Wb effect. We compare per-output-channel scales against group-wise scales (group sizes 128, 64). For closed-loop we built a three-process native harness (no Docker): the NeuroNCAP orchestrator, the NeuRAD renderer, and an OpenDriveVLA model node exposing inference plus an in-place re-quantization endpoint, so techniques are swept on the loaded model without reloading. The model node reconstructs the UniAD perception input (lidar2img, CAN-bus, ego-state prompt) from rendered frames each step. We report generation coherence: the fraction of frames on which the planner emits a well-formed (non-degenerate) plan.

IV. Results

A. Open-loop is blind

INT4 is lossless; the $19\times$ ego-ablation shows why the metric cannot reflect the perception/ planning pathway or its quantization.

a) On-device validation (Jetson AGX Orin 64 GB).: We replayed the captured planner inputs on a real AGX Orin (JetPack 6, torch 2.8) and re-scored open-loop L2 (40-sample mini subset, ST-P3). The Orin reproduces the A100 to ≈ 1 cm mean trajectory difference at FP16, and open-loop L2 is unchanged by quantization on-device: FP16 0.87 m vs group-INT4 0.90 m vs per-channel-INT4 0.88 m at 3 s. Yet the per-sample W4 trajectories already differ from FP16 by 14–24 cm (mean) on the Orin—a perturbation open-loop L2 averages away but closed-loop coherence exposes. The blindness is thus a property of the metric, not the hardware: it persists on the deployment target.

B. Closed-loop generation coherence

Across all 16 scenario instances per-channel INT4 collapses generation (mean 3.7% well-formed, $\leq 15\%$ on 15/16) while group-wise INT4 recovers most of FP16’s coherence (mean 73.5% vs 76.0%, $\geq 85\%$ of FP16 on 13/16). The same per-channel INT4 is lossless in-distribution (open-loop, Table 1), so the failure is a quantization $\times$ domain-shift interaction. Collision/NCAP is confounded and scene-dependent: on several scenarios every config including FP16 collides 10/10 (the base policy fails them), so only coherence cleanly separates the configs. We lead with coherence and do not claim group-wise exactly equals FP16 (it ties/beats on 13/16, partially recovers on 3: 0099, 0101, 0796).

C. Why granularity

A shared INT4 scale $s = \max|w|/7$ is set by the group’s largest weight; one outlier coarsens the grid for all weights it shares the scale with. Group-wise scales confine each outlier to its block, preserving precision on the signal-carrying weights. Under distribution shift the planner’s hidden states drift; the per-channel coarsened grid then tips greedy decoding into malformed trajectory text, whereas the group-wise grid preserves coherence—at a cost of $16/128 = 0.125$ bits/weight.

D. Memory benchmarks (toward edge)

The robust recipe (group-wise INT4) is ≈ 0.26 GB; the full inference node fits in 3–5 GB. Perception (FP32, custom CUDA ops) dominates wall-clock, so the quantization win is robustness, not latency; the edge bottleneck is perception, not the planner (see companion systems study).

V. Discussion

The two halves compose into one message: open-loop L2 hides quantization damage because it is ego-extrapolation, and closed-loop reveals it because perception/planning are on the critical path. The damage, when revealed, is a granularity artifact: per-channel INT4 is fine in-distribution but loses generation coherence under deployment shift, while group-wise INT4 is robust. Practitioners should (i) never certify a quantized driving policy on open-loop L2, and (ii) default to group-wise INT4 for the planner.

VI. Limitations and threats to validity

(1) Scene coverage (all 14 NeuRAD-reconstructed NeuroNCAP scenes, 16 scenario instances); this is the full released closed-loop benchmark, not all of nuScenes. (2) Out-of-distribution renderer: rendered inputs are OOD to real nuScenes—on several scenarios even FP16 fails the collision test (10/10)—so absolute NCAP/collision numbers are not comparable to in-distribution operation and our claims are relative (group $\approx$ FP16 $\gg$ per-channel on coherence). (2b) Partial recovery: group-wise ties/beats FP16 on 13/16 scenarios but only partially recovers on 3 (0099, 0101, 0796). (3) The collision metric is a confounded downstream proxy (collapse $\rightarrow$ frozen $\rightarrow$ geometry-dependent crash, mediated by the degenerate fallback); we therefore lead with coherence. (4) An activation-aware (AWQ) low-bit variant is excluded pending a fix. (5) Fake-quant (dequantized) weights are used, so the static planner footprint is measured on-device (Jetson AGX Orin 64 GB: 2.88 GB peak, 14.9 tok/s, group-wise INT4 packs the planner 0.72 $\rightarrow$ 0.19 GB) but a packed-INT4 speedup still requires a custom kernel and is not yet realized.

VII. Conclusion

For quantized autonomous-driving VLAs, open-loop L2 is blind ($19\times$ ego-ablation) and closed-loop is necessary: there, naive per-channel INT4 collapses generation under deployment-domain shift while group-wise INT4 (≈ 0.1 extra bits/weight) matches FP16. Use closed-loop coherence as the metric and group-wise INT4 as the recipe.

References

- [1] OpenDriveVLA. 2025. (Replace with verified citation.)
- [2] Hu et al. Planning-oriented Autonomous Driving (UniAD). CVPR 2023.
- [3] Zhai et al. Rethinking Open-Loop Evaluation of End-to-End AD. 2023.
- [4] NeuroNCAP. ECCV 2024.
- [5] NeuRAD: Neural Rendering for Autonomous Driving. CVPR 2024.
- [6] Lin et al. AWQ. MLSys 2024.
- [7] Frantar et al. GPTQ. ICLR 2023.

TABLE I
Open-loop nuScenes-mini ($n=162$). L2 in meters (ST-P3 protocol).

Setting	L2 (m)	well-formed
FP16	0.33	100%
INT8 / INT4 (per-channel)	0.33	100%
INT3	1.91	92%
INT2	6.45	0%
FP16, ego-state+history zeroed	6.38 ($19\times$)	100%

TABLE II
NeuroNCAP closed-loop generation coherence (well-formed predicted-frame rate), real-camera NeuRAD rendering, 14 scenes / 16 scenario instances, 10 paired seeds each. Per-scenario behaviour summarized; full per-scene table in the appendix/release.

Config	mean coherence	per-scenario behaviour
FP16	76.0%	—
INT4 per-channel	3.7%	collapses ($\leq 15\%$) on 15/16
INT4 group-128	73.5%	$\geq 85\%$ of FP16 on 13/16

TABLE III
Backbone weight footprint and overhead (357.8 M Linear params); effective bits include per-group FP16 scales.

Precision	eff. bits/weight	backbone weights
FP16	16.0	≈ 1.0 GB
INT4 per-channel	4.06	≈ 0.25 GB
INT4 group-128	4.125	≈ 0.26 GB